Predicting Coronary Heart Disease from BRFSS Survey Data

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Abstract—We predict whether a respondent reports myocardial infarction or coronary heart disease (MICHD) from 321 BRFSS survey variables [1]. Only 8.8% of the 328,135 training examples are positive, so we optimize F1 rather than accuracy. Our NumPy-only system combines questionnaire-aware preprocessing, categorical and missing-value indicators, nonlinear features, logistic regression, and a threshold fitted for F1. Stratified five-fold cross-validation improves from 0.3946 ± 0.0039 F1 for the MSE-gradient baseline to 0.4437 ± 0.0022. Training on all labelled rows yields 0.455 F1 and 0.882 accuracy on AIcrowd. The largest gains come from the decision threshold and preprocessing; regularization has little effect. We also identify a survey-design target proxy and discuss why the leaderboard result must not be interpreted as prospective clinical performance.

I. Problem and Data

The task is binary classification with labels $y \in \{-1, +1\}$, where $+1$ denotes MICHD. The inputs contain health, demographic, lifestyle, and survey administration variables. The strong class imbalance makes accuracy alone misleading: predicting only the majority class would be accurate but would detect no positive cases. We therefore use positive-class F1, $2TP/(2TP + FP + FN)$, as the model-selection metric and report accuracy as a secondary measure. Because the survey records existing diagnoses, our model predicts reported MICHD status, not the future causal risk of developing it.

II. Methods

A. Preprocessing and Features

All fitted preprocessing is learned only from the current training fold and then applied unchanged to its validation fold. Questionnaire codes from the 2015 BRFSS instrument [2] for "none", "don't know", and "refused" are converted to zero or missing as appropriate. Mixed units—for example kilograms versus pounds and weekly versus monthly frequencies—are converted to common units, and deterministic skip logic is resolved before imputation. Remaining missing values use training-fold medians, and every output column is standardized with training-fold statistics.

Categorical variables marked by our 321-column feature registry are one-hot encoded, with a separate unknown level. Missing indicators preserve whether a numeric value was imputed. We add squared standardized age and BMI, while 14 heavy-tailed count variables are clipped at the training 99.5th percentile and transformed with $\log(1 + x)$. Module-question flags retain information from sparse optional survey sections. Constant and more-than-95%-missing raw columns are otherwise removed.

Because a linear score cannot create interactions, we add 131 fixed columns: age and sex interactions with risk factors, pairwise risk-factor products, linear and squared counts of cardiovascular, chronic, and disability factors, and interactions of general health with age, sex, and risk factors. Seven further columns interact whether the optional daily-aspirin question was answered with age, sex, general health, high blood pressure, high cholesterol, diabetes, and blood-pressure medication. Together with the preceding transformations, the selected design matrix has 729 columns before these interactions and 736 columns in the final submitted design.

B. Model, Threshold, and Tuning

We implemented the six required regression methods from scratch using NumPy. The selected classifier is logistic regression. For labels converted to $\{0, 1\}$, it minimizes the mean binary cross-entropy

$$\mathcal{L}(w) = \frac{1}{N} \sum_{i=1}^{N} \left[\log(1 + e^{x_i^\top w}) - y_i x_i^\top w \right] \tag{1}$$

by full-batch gradient descent from $w = 0$. Scores $x^\top w$ are converted to labels using the cutoff that maximizes F1 on the training rows. This cutoff is refitted inside every fold, so validation labels are never used to choose it.

We searched step sizes $\gamma \in \{0.05, 0.08, 0.10, 0.12\}$ at 500, 1000, 2000, and 3000 iterations. At the selected optimizer setting we evaluated $\lambda \in \{0, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$. To avoid paying for negligible differences, we chose the cheapest and simplest setting within 0.0005 F1 of the best mean. This selected $\gamma = 0.10$, 1000 iterations, and $\lambda = 0$.

C. Evaluation Protocol

Every comparison uses the same five stratified folds and seed 42 on all 328,135 labelled rows. Each fold contains approximately 262,508 training and 65,627 validation rows. Imputation, scaling, clipping, encodings, model weights, and the threshold are fitted anew on its training partition. We report mean and population standard deviation across folds. The final AIcrowd file is produced by fitting the selected pipeline once on all labelled rows and predicting the 109,379 test rows.

TABLE I
Five-fold ablation on all training rows. Δ is relative to the preceding row, except the L2 and no-flags comparisons, which use the tuned optimizer row.

Configuration	CV F1	Δ
MSE GD + prevalence cutoff	$0.3946 \pm .0039$	—
Logistic + prevalence cutoff	$0.4023 \pm .0029$	+.0076
Logistic + F1 cutoff	$0.4219 \pm .0022$	+.0197
+ questionnaire-aware features	$0.4353 \pm .0022$	+.0134
+ tuned optimizer	$0.4394 \pm .0036$	+.0041
+ L2 ($\lambda = 10^{-3}$)	$0.4396 \pm .0032$	+.0002
+ engineered features	$0.4430 \pm .0026$	+.0035
+ aspirin interactions	$\mathbf{0.4437 \pm .0022}$	+.0007
Tuned model, no module flags	$0.4295 \pm .0037$	−.0099

TABLE II
Mean five-fold F1 across the optimizer grid.

$\gamma \backslash$ iterations	500	1000	2000	3000
.05	.4353	.4375	.4393	.4395
.08	.4371	.4386	.4391	.4391
.10	.4374	.4394	.4391	.4388
.12	.4377	.4393	.4391	.4387

III. Results

Table I shows a controlled chain. The first three rows give reference configurations: squared loss, logistic loss, and an F1-specific cutoff. The largest single gain is choosing the decision threshold for F1. The codebook-aware feature pipeline contributes a further 0.0134 F1, and optimizer tuning improves all five folds by another 0.0041. Engineered features add 0.0035 F1 in all five folds; aspirin interactions add 0.0007 in four of five folds, within the fold spread. We retain them because this frozen configuration gave the best AIcrowd score. The complete system is 0.0490 F1 above the original baseline.

Convergence explains the last gain. At $\gamma = 0.05$, increasing from 500 to 2000 iterations raises F1 from 0.4353 to 0.4393, after which improvement is negligible. Doubling the step size reaches the same region in 1000 iterations. The largest regularized mean, 0.4396 at $\lambda = 10^{-3}$, differs from the unregularized 0.4394 by only 0.0002; $\lambda = 10^{-2}$ reduces F1 to 0.4369. We therefore retain the simpler unregularized model.

The initial uploaded logistic model scored 0.418 F1 and 0.897 accuracy on AIcrowd. The final system predicts 12.90% of test rows as positive and scores 0.455 F1 with 0.882 accuracy. The lower accuracy is expected: an F1-oriented cutoff predicts more positives, trading some majority-class accuracy for better detection of the minority class. The final AIcrowd F1 is 0.0113 above its CV estimate, which is compatible with sampling variation and differences between the public test set and our folds.

IV. Limitations and Discussion

Threshold selection and semantically correct preprocessing matter more here than regularization. However, one useful feature is also a serious limitation. HAREHAB1 asks about rehabilitation following a heart attack and is only shown to respondents reporting one; all 773 training respondents for whom it is answered are positive. A separate ablation estimated that sparse module flags, dominated by this variable, add about 0.0099 F1. The feature is present in both provided train and test data, but it is a direct target proxy, a recognized form of data leakage [3]. We retain the module flags and aspirin interactions to reproduce the benchmark submission; removing the module flags lowers CV F1 from 0.4394 to 0.4295. Consequently, this system is suitable for the defined benchmark, not for prospective screening. A clinically meaningful follow-up should remove all post-outcome questions and report the resulting performance separately.

The same folds were used for hyperparameter selection and the final CV number, so the reported mean may contain modest selection optimism. The independent AIcrowd score provides an additional check, but repeated leaderboard queries must not replace local validation. Nested CV or a sealed holdout would give a cleaner final generalization estimate.

V. Conclusion and Reproducibility

A transparent NumPy pipeline combining logistic loss, an F1-specific cutoff, and questionnaire-aware feature engineering reaches 0.455 AIcrowd F1. The ablation shows that the threshold and feature semantics account for most of the improvement, while optimizer tuning provides a smaller consistent gain and regularization is unnecessary at the tested scale.

The hand-in repository is https://github.com/Guggi99/ml-project1-2026-submission. Experiments share one cross-validation pipeline and store their folds, seed, transformations, model, and cutoff rule. With the three course CSV files in data/, python run.py –data-dir data –submission submission.csv reproduces the frozen submission (SHA-256 prefix 0a97469b), which scored 0.455 F1 and 0.882 accuracy on AIcrowd versus a CV estimate of 0.4437. Every push runs unit, formatting, and course grading tests. AI assistants were used in this project in compliance with EPFL rules. All design decisions, code and text were reviewed and are the responsibility of the authors.

References

[1] Centers for Disease Control and Prevention, "2015 BRFSS survey data and documentation," 2015. [Online]. Available: https://www.cdc.gov/brfss/annual_data/annual_2015.html

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[3] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," ACM Trans. Knowl. Discov. Data, vol. 6, no. 4, pp. 15:1–15:21, 2012, doi: 10.1145/2382577.2382579.